Course Title: Basics of Generative AI with Application to Chemical and Biological Physics

Start Date: August 2026 (Autumn Semester, August–December 2026)

Coordinates (Preferred time slots): Two 1.5 hour session per week — one 90-minute lecture and one 90-minute combined tutorial and in-class hands-on practice

No. of contact classes: Approximately 36 (12 teaching modules, each comprising a lecture, a tutorial, and a hands-on practice session over a full semester).

Instructor(s): Prof. Jagannath Mondal (TIFR Hyderabad); (+ a Teaching Assistant he will recommend)

Syllabus:
Each module pairs a lecture with an in-class tutorial and a hands-on coding/practice session. The course builds up gradually, from mathematical foundations and classical models to modern generative methods, closing with case studies drawn from contemporary research topics in chemical and biological physics.

1. **Mathematical and Numerical Preliminaries** – Essentials of linear algebra, probability and statistics; distributions, gradients, and the maximum-likelihood and optimization viewpoint that underlies machine learning. Basic Primer on Python and basic libraries
2. **Foundations of Machine Learning** – The learning problem cast as distribution and density estimation; supervised and unsupervised settings, loss functions, generalization, and the bias–variance trade-off.
3. **Regression and Classification** – Linear and logistic regression, regularization, model evaluation and metrics, and the probabilistic interpretation of fitting and prediction.
4. **Artificial Neural Networks**– From the perceptron to multilayer networks; backpropagation, activation functions, regularization and normalization, implemented without high-level libraries.
5. **Latent-Variable Models: Autoencoders and VAEs** – Representation learning, the encoder–decoder framework, the variational objective (ELBO), and the reparameterization trick for generative sampling.
6. **Generative Adversarial Networks (GANs)** – Adversarial training of generator and discriminator, characteristic failure modes such as mode collapse, and strategies for stable training.
7. **Sequence Models: From RNNs to Transformers** – Recurrent networks (RNN, LSTM, GRU), the attention mechanism, the transformer architecture, and autoregressive (GPT-style) generation.
8. **Diffusion and Flow-Based Models** – Denoising diffusion (DDPM), the score-based formulation, and normalizing and rectified flows for continuous generative modelling.
9. **Introduction to Reinforcement Learning (time permitting)** – Agents, rewards, policy- and value-based methods, and connections between reinforcement learning and generative modelling.
10. **Case Study I — Machine Learning for Molecular-Dynamics Sampling** – Dimensionality reduction, collective-variable discovery, and generative models applied to conformational landscapes and enhanced sampling.
11. **Case Study II — Classification in Chemical and Biological Physics** – Building, training, and benchmarking classifiers on simulation- and structure-derived data.
12. **Applications in Bioinformatics** – Regression for protein binding-affinity prediction, post-translational-modification (PTM) site prediction, and an introduction to protein language models (PLMs).

The semester includes a mid-term examination at its midpoint and concludes with a final project, written report, and presentation.

Pre-requisite courses (the courses which have to be taken before opting for this course):
Working knowledge of linear algebra, probability and statistics, and programming in Python. No prior background in machine learning is assumed. Familiarity with basic chemical or biological physics is helpful for the application modules but is not required.

Primary Text / Reference Books:

- I. Goodfellow, Y. Bengio and A. Courville, Deep Learning, MIT Press (2016).
- K. P. Murphy, Probabilistic Machine Learning: An Introduction and Advanced Topics, MIT Press (2022–2023).
- C. M. Bishop and H. Bishop, Deep Learning: Foundations and Concepts, Springer (2024).
- S. J. D. Prince, Understanding Deep Learning, MIT Press (2023).
- Selected reviews and primary research papers from research articles.

Evaluation Method (Weightage for Internal Assessment, Mid Term / Term End exams, Presentations etc.):

- Continuous internal assessment (in-class tutorials and hands-on assignments): 30%
- Mid-term examination: 30%
- Final project, written report, and presentation: 40%

Course Outcome (CO):

Sr. No.	On completing the course, the student will be able to:
CO 1	Formulate machine learning as a distribution- and density-estimation problem and apply the probabilistic, maximum-likelihood framework.
CO 2	Implement regression, classification, and a neural network from first principles, including backpropagation, regularization, and normalization.
CO 3	Construct, train, and interpret latent-variable generative models (autoencoders and variational autoencoders).
CO 4	Train generative adversarial networks and diagnose and mitigate common training pathologies.
CO 5	Analyse sequence models ranging from recurrent networks to attention-based transformers and autoregressive (GPT-style) generation.
CO 6	Derive and implement diffusion (DDPM) and flow-based generative models.
CO 7	Apply machine-learning and generative methods to molecular-dynamics sampling and classification problems in chemical and biological physics.
CO 8	Develop regression and prediction pipelines for bioinformatics tasks such as binding-affinity prediction, PTM-site prediction, and protein language models.